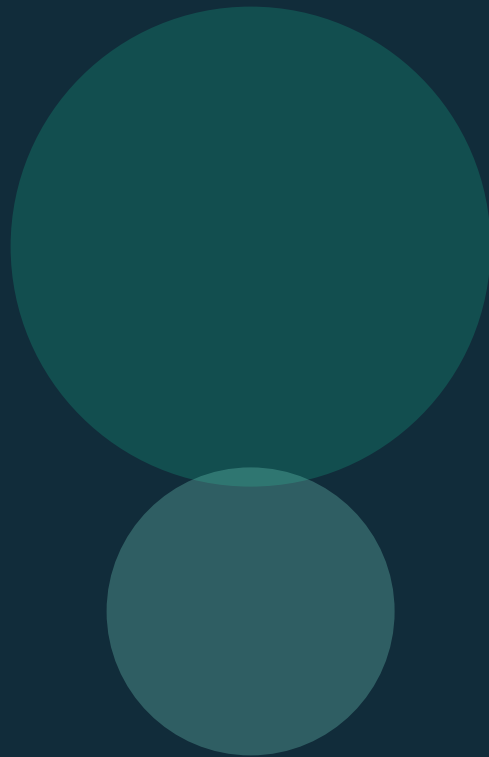


PALANTIR FOUNDRY

Connect to Data

Data Connection — Syncing, Sources, Runtimes & Capabilities



What is Data Connection?

An application used to synchronize data from external systems into Foundry — and to enable writeback to external systems via Webhooks and data exports.

Robustness

Automatic retries, small-batch syncs with low-complexity queries, integrated data health monitoring, and an "as-is" ingestion philosophy to keep raw data untransformed.

Extensibility

Out-of-the-box support for databases, FTPS, HDFS, S3, SFTP, and local directories, plus flexibility to connect to new system types using existing plugins.

Ease of Use

Backend services handle the bulk of the work — scheduling, auth, orchestration, and monitoring — exposed through a simple frontend UI for all users.

Core Concepts: Sources & Credentials

Sources

A **Source** represents a single connection to a target system, including configuration and credentials.

Connectors (Source Types):

A broad range of connectors support the most common data systems. For systems without a dedicated connector, the generic connector or REST API source may be used.

Runtime: Every source must be assigned a runtime, which defines networking configuration and where capabilities execute.

Credentials

- Secret values required to access a source system (passwords, tokens, API keys).
- All credentials are encrypted and stored securely in the platform.
- Stored locally on an agent, or directly in the platform, depending on runtime.
- Some sources authenticate without secrets (OpenID Connect, outbound apps, cloud identity).

Runtimes

Three options define how Foundry connects to your systems:

Direct Connection

RECOMMENDED

Preferred when data source is accessible over the Internet. No agent needed — avoids operational overhead.

Networking:

Target system must allow inbound traffic from Palantir egress IPs.

Execution:

Capabilities run in Foundry.

Agent Proxy

Used when target is not reachable from the Internet. Functions almost identically to direct connection.

Networking:

Agent on your infrastructure reverse-proxies traffic to unreachable systems.

Execution:

Capabilities run in Foundry.

Agent Worker

FALLBACK

Use only when the connector does not support agent proxy. Configuration and credentials stored locally on agent.

Networking:

Agent on your infrastructure runs jobs and pushes/pulls data from Foundry.

Execution:

Capabilities run on customer Linux host.

Source Capabilities

Inbound (Data into Foundry)

Batch Syncs

Read from external system → write to Foundry dataset (file or table)

Streaming Syncs

Low-latency feeds from Kafka, Kinesis, Google Pub/Sub

CDC Syncs

Change data capture with changelog metadata for database replication

Media Syncs

Import media data (PDFs, images, video) into media sets

HyperAuto

Auto-discover SAP schema and automate syncs & ontology creation

Outbound & Other

File Exports

Push files from a Foundry dataset to an external filesystem

Table Exports

Export rows with schema from a dataset to an external database

Streaming Exports

Export data from a Foundry stream to an external queue/topic

Webhooks

Make structured interactive requests to an external system

Virtual Tables/Media

Register external data without copying it into Foundry

Exploration

Interactively explore an external system before syncing

Batch & Streaming Syncs

Batch Syncs

Read data from an external system and write it into a Foundry dataset. Can be configured incrementally, with or without a schema.

File Batch Syncs

Sync files without a schema. Common systems: S3, ABFS, Google Cloud Storage, SMB, SharePoint Online.

Table Batch Syncs

Sync data with a schema, including type translation. Common systems: SQL Server, Postgres, SAP, Salesforce, NetSuite.

Streaming Syncs

Stream data from low-latency feeds into a streaming dataset. Supported systems include Kafka, Amazon Kinesis, and Google Pub/Sub.

Change Data Capture (CDC)

Like streaming syncs, but with additional changelog metadata automatically propagated. Used for databases that support low-latency replication.

Initial Setup Overview

1 Connection

- Agent — for private networks / on-premises
- Direct Connection — for Internet-accessible sources (preferred)

2 Source / Connector

- Configure the source type (Postgres, S3, SAP, REST API, etc.)
- Assign credentials and runtime to the source

3 Sync

- Define what data to read and which dataset to output
- Once run, result is a Foundry dataset ready for use

Setting Up a Direct Connection

1

Configure Network Egress Policy

In Control Panel, navigate to Network Egress. Add a policy with DNS hostname (for HTTPS) or CIDR + port. Requires Information Security Officer role.

2

Provision Credentials

Create a dedicated service account for Foundry. Best practice is to use a service account following internal guidelines. Note credentials for next step.

3

Create Source in Data Connection

Navigate to Data Connection → Sources → New source. Select source type, choose Direct Connection, then Continue.

4

Save & Choose Network Policy

Name the Source, save to a Project (one per Source recommended). Select the network egress policy configured in step 1.

5

Add Credentials & Save

Add the provisioned credentials, configure source-specific details (URLs, regions, drivers for JDBC). Save to complete setup.

Agent Setup & High Availability

Agent Hardware Requirements

OS: 64-bit Linux (RHEL 8, Ubuntu 22.04 or equivalent)

CPU: 4 cores

RAM: 16 GB

Storage: 500 GB free at /opt (SSD preferred)

Note: One agent per host. Running multiple agents on the same machine is not supported.

High Availability Best Practices

- Install agents in pairs on similar hardware.
- Use paired naming: agent-1 and agent-2.
- Assign both agents to every source.
- Configure non-overlapping upgrade windows for each pair.
- HA is at the source level — Foundry dispatches to a healthy agent automatically.
- Add extra agents at source creation time; re-adding later requires re-entering credentials.

Key Takeaways

01

Data Connection centralizes all sync workflows —

Inbound syncs, outbound exports, webhooks, virtual tables — all managed from one application.

02

Three core principles guide the design —

Robustness (auto-retry, as-is ingestion), Extensibility (broad connector support), and Ease of use (UI-driven, less IT burden).

03

Choose the right runtime —

Direct connection (preferred for Internet-accessible sources), Agent Proxy, or Agent Worker — in that priority order.

04

Setup follows a 3-step sequence —

Connection → Source/Connector → Sync. Agent setup is rare; syncs are configured frequently.

05

High availability is built in —

Deploy agents in pairs with non-overlapping maintenance windows. Foundry automatically routes to healthy agents.